Formal Logic 1

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I. **Week 1: What Is Logic?**

Reading: Magnus et al., 2025, Chapters 1-3 (pp. 2-25)

Focus: validity, soundness, entailment, satisfiability; distinguishing deductive from inductive arguments

A. Arguments
B. The scope of logic
C. Other logical notions

II. **Week 2: Symbolizing English**

Reading: Magnus et al., 2025, Chapters 4-5 (pp. 27-48)

Focus: translating natural language into TFL; understanding connective meanings

A. First steps to symbolization
B. Connectives

III. **Week 3: TFL Syntax**

Reading: Magnus et al., 2025, Chapters 6-8 (pp. 49-67)

Focus: well-formed formulas, parse trees, scope, avoiding ambiguity

A. Sentences of TFL
B. Ambiguity
C. Use and mention

IV. **Week 4: TFL Semantics**

Reading: Magnus et al., 2025, Chapters 9-11 (pp. 69-84)

Focus: understanding truth-functional connectives; constructing truth tables for complex sentences;

A. Characteristic truth tables
B. Truth-functional connectives
C. Complete truth tables

V. **Week 5: Semantic Concepts**

Reading: Magnus et al., 2025, Chapters 12-14 (pp. 85-101)

Focus: using truth tables to test logical properties; efficiency strategies

A. Semantic concepts
B. Limitations of TFL
C. Truth table shortcuts

VI. **Week 6: Partial Truth Tables**

Reading: Magnus et al., 2025, Chapter 15 (pp. 102-8)

Focus: using partial tables for validation; catch-up and review

A. Partial Truth Tables

VII. **Week 7: Introduction to Natural Deduction**

Focus: the Fitch-style proof system; introduction and elimination rules

A. The very idea of natural deduction
    1. Magnus et al., 2025, Chapter 16 (pp 110-2)
B. Basic rules for TFL
    1. Magnus et al., 2025, Chapter 17 (pp 113-41)

VIII. **Week 8: Constructing Proofs**

Focus: proof strategies; working backward from the conclusion; subproofs

A. Constructing proofs
    1. Magnus et al., 2025, Chapter 18 (pp 142-161)

IX. **Week 9: Additional Rules & Proof-Theoretic Concepts**

Focus: extending the basic system; connecting proof theory to semantics

A. Additional rules for TFL
    1. Magnus et al., 2025, Chapter 19 (pp 162-9)
B. Proof-theoretic concepts
    1. Magnus et al., 2025, Chapter 20 (pp 170-3)

C. Derived rules

1. Magnus et al., 2025, Chapter 21 (pp 174-81)

X. **Week 10: Soundness, Completeness**

Focus: relating semantic and proof-theoretic consequence; reviewing TFL natural deduction

A. Soundness and completeness

1. Magnus et al., 2025, Chapter 22 (pp 182-8)

XI. **Week 11: Building Blocks of FOL**

Focus: translating simple universal and existential claims; predicate notation

A. Building Blocks of FOL

1. Magnus et al., 2025, Chapter 23 (pp 190-8)

B. Sentences with One Quantifier

1. Magnus et al., 2025, Chapter 24 (pp 199-212)

XII. **Week 12: Multiple Generality & Identity**

Focus: nested quantifiers; the identity predicate; definite descriptions

A. Multiple generality

1. Magnus et al., 2025, Chapter 25 (pp 213-27)

B. Identity

1. Magnus et al., 2025, Chapter 26 (pp 228-34)

C. Sentences of FOL

1. Magnus et al., 2025, Chapter 27 (pp235-42)

XIII. **Week 13: Definite Descriptions & FOL Ambiguity**

Focus: Russellian analysis of "the"; scope ambiguities in FOL

A. Definite descriptions

1. Magnus et al., 2025, Chapter 28 (pp 243-52)

B. Ambiguity

1. Magnus et al., 2025, Chapter 29 (pp 253-6)

XIV. **Week 14: FOL Semantics**

Focus: defining truth relative to an interpretation; testing validity via counter-models

A. Extensionality

1. Magnus et al., 2025, Chapter 30 (pp 258-66)

B. Truth in FOL
   1. Magnus et al., 2025, Chapter 31 (pp 267-76)

C. Semantic concepts
   1. Magnus et al., 2025, Chapter 32 (pp 277-80)

D. Using interpretations
   1. Magnus et al., 2025, Chapter 33 (pp 281-8)

E. Reasoning about interpretations
   1. Magnus et al., 2025, Chapter 34 (pp 289-92)

F. ** Properties of relations
   1. Magnus et al., 2025, Chapter 35 (pp 293-5)

XV. **Week 15: FOL Proofs**

Focus: universal and existential introduction/elimination; identity rules; connecting FOL proof theory to semantics

A. Basic rules for FOL
   1. Magnus et al., 2025, Chapter 36 (pp 297-315)

B. Proofs with quantifiers
   1. Magnus et al., 2025, Chapter 37 (pp 316-22)

C. Conversion of quantifiers
   1. Magnus et al., 2025, Chapter 38 (pp 323-5)

D. Rules for identity
   1. Magnus et al., 2025, Chapter 39 (pp 326-9)

E. ** Derived rules
   1. Magnus et al., 2025, Chapter 40 (pp 330-1)

F. ** Proofs and semantics
   1. Magnus et al., 2025, Chapter 41 (pp 332-5)

References

Magnus, P., Button, T., Trueman, R., & Zach, R. (2025). *Forall x: Calgary: An introduction to formal logic*. University of Calgary. https://openlogicproject.org/